Mohammad Uzair Khan

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Education

BITS Pilani, Dubai Campus — Dubai, UAE
B.E. Computer Science — *Expected 2027*
- Relevant Coursework: Database Management Systems, Machine Learning, Data Structures & Algorithms, Large Language Models

Bishop Conrad Senior Secondary School | *Senior Secondary (Class XII)* — Bareilly, India | *2023*

Data Science & AI Projects

PharmaEvidence: evidence-grounded claim review | *Python, BM25, DeBERTa NLI, ONNX, Next.js* — Live | Code
- Parsed four versioned DailyMed labels (KEYTRUDA, WINREVAIR) into **2,309** evidence passages; combined BM25 retrieval, numeric and qualifier rules and a pretrained **INT8 DeBERTa NLI** model running in the browser
- Froze **120** authored claims split by fact family (72 development / 48 held out): held out, rules scored 40/48, NLI **47/48**, rules + NLI **48/48**; the rules caught a tenfold dose error the model accepted

Rx Market Pulse: forecasting and an AI analyst | *Python, SQL (DuckDB), TypeScript, LLM tool calling* — Live | Code
- Backtested quarterly spend forecasts over **28 quarters**, choosing on 2022–23 and testing on 2024–25: the chosen method scored 9.01% WAPE, worse than the seasonal baseline's **5.54%**, reported as found
- Built an analyst agent over **36.1M** Medicaid rows with **five typed tools** (spend, growth drivers, territories, forecasts, trials), provenance and refusals; passed 20/20 curated real-model smoke checks

Footnote: question answering over company filings | *Python, BM25, embeddings, LLM agents, Next.js* — Live | Code
- Measured retrieval on all **150** FinanceBench questions over 360 filings (53,399 pages): hit@10 **44%** with the right filing, 13% across all, 31% with a company/year filter; searching the gold text hits **97%**, so the gap is vocabulary
- Built cited answers with a grounding check, a whitelisted calculator, an agent and a model judge cross-checked by a rule, with bootstrap intervals

SignalDesk AI: oncology competitive intelligence | *Python, SQL, Gemini, TypeScript* — Live | Code
- Linked **29** FDA announcements, 8 formulations and **30** trial records; a bounded Gemini agent with five typed tools and citation checks, deployed with Redis-backed quotas
- Paired evaluation, 20 development / **40 held-out** tasks against a same-model BM25 baseline: agent 17/40 vs baseline 19/40, short of the 34/40 target; published case-level results and failure analysis

Field Force Planner: territory alignment from the NPI registry | *Python, SQL (DuckDB), Excel* — Live | Code
- Sized a launch from **9.9M** registry records (1,310 diabetes reps) and cut 3,143 counties into connected territories by weighted k-means seeds and local search: spread **7.5%** vs 274% by county count; prevalence tracks state spend by size (Spearman 0.86) but not per enrollee (–0.10)

Also built: Drug Price Watch *(chained price index, 17.3M prices)* · Payer Access Intelligence · Trial Pipeline Benchmark — GitHub

Experience

Backend Developer Intern, Amaani (Travel-Tech Startup) — Dubai, UAE
BITS Practice School (PS-1) — *Jun 2025 – Aug 2025*
- Developed and maintained backend services in **Python (Flask)**, building RESTful APIs for core travel-booking features
- Documented API contracts with **Swagger/OpenAPI** and validated endpoints end-to-end using **Postman**; implemented request validation and data models

Research & Publications

Lightweight Deep Learning for Tomato Leaf Disease Detection on Edge Devices — 2025 – 2026
First Author; submitted to IEEE MSN 2026 (under review)
- Trained a MobileNetV2 classifier (10 classes) reaching **97.33% test accuracy** (macro-F1 0.963); compressed it to **5.3 MB** with INT8 quantization and ran it on a Raspberry Pi 4B at about 250 ms per image

Technical Skills

Machine Learning & AI: scikit-learn, TensorFlow/Keras, TensorFlow Lite, INT8 quantization, Transformers.js, ONNX Runtime, NLI, retrieval (BM25, embeddings), RAG, LLM agents and tool calling (Claude, Gemini)
Statistics & Evaluation: held-out and paired evaluation, bootstrap intervals, backtesting (WAPE), rank correlation, price indexes, optimisation heuristics
Data & Engineering: Python (Pandas, NumPy), SQL (DuckDB, MySQL), Excel, Power BI (DAX), TypeScript, Flask, REST APIs, Next.js/React
Tools: Git, GitHub Actions, pytest, Playwright, Postman, LaTeX

Certifications

The Ultimate Job Ready Data Science Course – CodeWithHarry (2026)